Isha Sharma
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EDUCATION

- 2018-2024 : PhD, Physics, Indian Institute of Technology Kharagpur, West Bengal, India.
- 2016-2018 : Masters of Science, Physics, Indian Institute of Technology Kharagpur, West Bengal, India.
  CGPA: 7.89.
- 2013-2016 : Bachelor of Science, Physics, Miranda House, Delhi University, Delhi, India.
  CGPA: 8.96
- 2013 : Class 12, D.A.V. Public School, Mahendragarh, Haryana, India.
  Percentage: 92%
- 2011 : Class 10, D.A.V. Public School, Mahendragarh, Haryana, India.
  CGPA: 10

THESIS

- Ph.D thesis : Technology and Modelling of Fiber-Optic Cantilever Deflection Device for Electric Field Sensing and Mode Switching Applications
- M.Sc. project : Protein-Protein Interaction through Light Scattering Technique
  The molecular dimension of the protein is determined in terms of Stokes radius using light scattering techniques.

AREA OF RESEARCH

- Fiber and integrated optics, photonics, and optical fiber sensors and devices.
- Designing, fabricating, and optimizing optical setups, with a specific emphasis on optical fiber devices.
- Multiferroic materials for the development of all-fiber electric and magnetic field sensors.

LIST OF PUBLICATIONS (7)

- **I. Sharma**, P. Roy Chaudhuri, "Demonstration of Mode-Switching in a Few-Mode Fiber using Electric Field Controlled Dynamic Off-Set Coupling," *Journal of Lightwave Technology*, doi: 10.1109/JLT.2024.3370846 (2024).
- **I. Sharma**, P. Roy Chaudhuri, "Electric field sensing and polarisation measurement using advanced multi-pass interrogation type fiber-optic beam deflection probe," *Optical Fiber Technology*, vol. 81, p. 103484, doi:10.1016/j.yofte.2023.103484 (2023).
- **I. Sharma** , P. Roy Chaudhuri , "A new approach to sensing low electric field using optical fibers beam-deflection configuration with $BiFe_{0.9}Co_{0.1}O_3$ nanoparticles as probe and determination of polarisation," , *Optical Fiber Technology*, vol. 62, pp. 1-8, doi:10.1016/j.yofte.2021.102472 (2021).
- P. Roy Chaudhuri , **I. Sharma** , "Determination of polarization properties of piezoelectric nanocomposite particles $BiFe_{0.9}Co_{0.1}O_3$ using fiber-optic cantilever beam deflection approach," *Journal of Optics*, vol. 50(4), pp. 611-620, doi:10.1007/s12596-021-00741-8 (2021).
- **I. Sharma** , P. Roy Chaudhuri , "Design, experimental demonstration and modeling fiber ring resonator device with fiber cantilever-deflection probe for sensing surrounding electric field," *A. Journal of Physics*, vol. 31(3-6), pp.A27-A32, doi: 10.54955/AJP.31.3-6.2022.A27-A32 (2022).

MANUSCRIPT FOR SUBMISSION

- A comprehensive review of fiber-optic cantilever beam deflection devices for electric and magnetic sensing applications.

CONFERENCES PROCEEDINGS/ BOOK CHAPTERS (10)

- **I. Sharma** , P. Roy Chaudhuri, "Demonstration of precise mode switching in four mode fiber using electric field induced launching offset control" *In Frontiers in Optics*, Optica Publishing Group, pp. JTu4A-66, doi: 10.1364/FIO.2023.JTu4A.66 (2023).
- **I. Sharma**, P. Roy Chaudhuri, "Experimental Demonstration of All-Fiber Electric Field Sensing Device" *International Symposium on Devices, Circuits and Systems (ISDCS)*, vol. 1, pp. 1-3, IEEE (2023).
- **I. Sharma**, P. Roy Chaudhuri, "Experimental Demonstration of Electric Field Sensing Using Sagnac Loop Based Fiber Cantilever Configuration" 5th International Symposium on Devices, Circuits and Systems (ISDCS). IEEE, doi: 10.1007/978-981-99-0055-826 (2022).

AWARDS AND FELLOWSHIPS

- National Eligibility Test-Junior Research Fellow (NET-JRF) (CSIR) - June 2018
- Graduate Aptitude Test in Engineering (GATE) in Physics - 2018
- Indian Institute of Technology-Joint Admission Test (IIT-JAM) in Physics - 2016
- Awarded 'CSSS Scholarship' offered by MHRD, Govt. of India 2013-2018
- Cleared Olympiads in Mathematics (Class XI) organised by Science Olympiad Foundation - 2012
- Selected as Guest faculty at Shiva ji College, Delhi University

COMPUTER ANALYSIS AND SIMULATION

- C++, Matlab, COMSOL Multiphysics.

TEACHING EXPERIENCE

- Electromagnetism
- Mathematical Physics
- Physics of Semiconductor
- Fiber optics and Optoelectronics
- Quantum Mechanics
- Physics of Waves
- Btech Ist year Lab

DETAILS OF REFEREES

- Prof. Partha Roy Chaudhuri

  Department of Physics

  Indian Institute of Technology, Kharagpur

  Kharagpur, West Bengal, India - 721 302

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- Prof. Maruthi Manoj Brundavanam
  Department of Physics
  Indian Institute of Technology, Kharagpur
  Kharagpur, West Bengal, India - 721 302
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- Prof. Abha Dev Habib
  Department of Physics
  Miranda House, Univeristy of Delhi
  New Delhi, India - 110 0007
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